HEALTHCARE PERFORMANCE DATASET CLEANING PROJECT REPORT

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OBJECTIVE

The objective of this data cleaning process was to ensure data integrity by resolving logical inconsistencies, missing values, and structural issues, and to prepare the dataset for further analysis. DATA CLEANING was done using Microsoft Excel.

DATASET STRUCTURE OVERVIEW

- File type: CSV
- The dataset contains 5,000 encounter records.
- 989 unique patient IDs
- 17 key variables
- 8 ICD-10 diagnosis codes: B34, E11, I10, K35, M54, N39, J18, A09
- 6 unique CPT procedures: 93000, 70450, 80053, 99284, 81002, 71020
- 6 departments namely: Emergency, Pediatrics, Orthopedics, General Surgery, Cardiology, and Internal Medicine

KEY DATA QUALITY ISSUES IDENTIFIED, AND CLEANING STEPS/APPROACH

DATA CLEANING TOOL: Microsoft Excel

I. **Inconsistent data types** (reformatted):
Patient_ID → GENERAL → TEXT
Encounter_ID → GENERAL → TEXT
Admission_Datetime → GENERAL → DATE/TIME
Discharge_Datetime → GENERAL → DATE/TIME
ed_arrival_time → GENERAL → DATE/TIME
provider_seen_time → GENERAL → DATE/TIME
icd10_diagnosis → GENERAL → TEXT
cpt_procedure → GENERAL → TEXT
cost_of_encounter → GENERAL → ACCOUNTING
Outcome → GENERAL → TEXT
Age → GENERAL → NUMERIC
Gender → GENERAL → TEXT
Department → GENERAL → TEXT
Admission_type → GENERAL → TEXT
Discharge_Disposition → GENERAL → TEXT
severity_level → GENERAL → NUMERIC

readmission_flag → GENERAL → NUMERIC

II. **Inconsistent Categorical Values**: Some categorical columns contained inconsistent capitalization. E.G. Admission Type column: “Emergency”, “emergency”

III. **None values in Gender Column**: The Gender column contained 150 entries recorded as “None”, which does not represent a valid gender category.

IV. **Department and Age Mismatch**: The Pediatrics department contained patients with ages above 18, which contradicts standard clinical classification, where pediatric patients are typically under 18 years.

V. **Multiple Ages for One Patient:** Some patients had different ages recorded across encounters

VI. **Blank cells in Cost of Encounter Column:** The Cost of Encounter column contained missing values, which could distort calculations related to financial metrics.

VII. **Logical Contradictions Between Admission Type and Department:** Certain Patients assigned to the Emergency department but having Admission Type listed as Elective.

VIII. **Outcome and discharge disposition conflicts:** Some records showed inconsistencies between Outcome and Discharge Disposition. Some record shows Outcome recorded as Expired, but disposition listed as Home, and some, Discharge_Disposition marked as 'Expired' while Outcome was recorded as 'Recovered'

IX. **RE-ADMISSION flag and outcome conflict:** Some records shows contradicting readmission_flag, and outcome value. Some re-admitted patients had re-admission flag as “0”.

X. ED arrival time for patients with Elective admission types

CLEANING PRINCIPLES AND STEPS APPLIED

I. Standardized all cases using the proper function.

II. Removed excess spaces using the TRIM function.

III. Formatted columns from General to TEXT, NUMERIC, and DATE/TIME as required

IV. **GENDER AND ID CONFLICT**: Pivots table was created to summarize the number of Males and Females in each ID, and using the IF and GET function { =IF(GETPIVOTDATA("gender", Sheet1!$A$3, "patient_id", A2, "gender", "Male") > GETPIVOTDATA("gender", Sheet1!$A$3, "patient_id", A2, "gender", "Female"), "Male", "Female")}. The gender was filled in using the highest number of gender for each ID.

V. **DEPARTMENT AND AGE MISMATCHED**: This was calculated using 18 years as the age limit for supposed patients in Pediatrics. Using the IF function =IF(AND(K2>18,M2="Pediatrics"),"Misclassified",M2), Patients above 18yrs were flagged and labelled as Misclassified.

VI. **Resolving Multiple Ages for One Patient:** The median age per patient was calculated. This value was applied consistently to that patient's records.

VII. **BLANK CELLS IN COST OF ENCOUNTER COLUMN:** Using the Go To Special (Ctrl+G) tool in excel, blank cells in the cost of encounter column was highlighted. This was then filled with the median of the cost of encounter value, to prevent skewing of the distribution.

VIII. **ELECTIVE ADMISSION TYPE IN EMERGENCY DEPARTMENT:** This was resolved using the function =IF(O2="Emergency","Emergency",P2) i.e =IF(department= Emergency, emergency, ELSE Admission type)

IX. **RECALCULATION OF READMISSION FLAG**: The original readmission flag was unreliable, so a new one was calculated using the 30-day readmission rule.

The dataset was sorted by: Patient id, and Admission date columns

Previous discharge date column D was created using =IFERROR(MAXIFS(F$2:F2, A$2:A2,A2,C$2:C2,"<"&C2,F$2:F2,"<"&C2)," ")

Days since last discharge column E was calculated using =IF(D2=" ", " ", C2-D2)

Corrected readmission flag was created using = IF(E2<0, "Invalid", IF(AND(E2>0, E2<=30),1,0)). 1 if patient returned within 30 days, 0 if admission occurred after 30 days.

This ensured that readmission metrics were based on actual admission intervals rather than unreliable flags.

X. **OUTCOME AND DISCHARGE DISPOSITION CONFLICTS**: First the outcome column was calculated using the readmission flag =IF(T2=1,"Readmitted",M2). T2(readmittion flag) M2=OUTCOME. Then, =IF(Discharge_Disposition = 'Expired', then Outcome = 'Expired'; otherwise retain original Outcome).

XI. **ED ARRIVAL TIME FOR PATIENTS WITH ELECTIVE ADMISSION TYPES:** This was calculated using the formula =IF(Q2="Elective", " ",F2). Affected cells were left blank.

XII. Improving Interpretability of Diagnosis and Procedure Codes: Two lookup tables were created to convert codes into readable descriptions.

ICD_Code	ICD_Readable
B34	Viral Infection
E11	DM ii
I10	Hypertension.p
K35	Appendicitis.a
M54	Dorsalgia
N39	UTI.u
J18	Pneumonia.u
A09	Gastroenteritis

CPT_Code	CPT_Readable
93000	ECG
70450	NCT.b
80053	Lab test
99284	ED visit
81002	Urinalysis
71020	Chest XR

Final Dataset Preparation

After cleaning was completed, A new sheet, **Clean_Data,** was created. The cleaned columns were used to replace the original columns where necessary, and new columns were added as necessary.

CONCLUSION

At first glance, the dataset appeared fairly structured, with 5,000 encounter records and multiple variables. However, upon examination, several inconsistencies and conflicts became clear. This ranged from simple formatting inconsistencies to deeper structural problems that could significantly affect how hospital performance metrics are interpreted.

Some of the cleaning tasks were straightforward. For example, standardizing text fields helped resolve inconsistencies such as different capitalization in department and admission type categories. These fixes ensured that categories would not be treated as separate groups during analysis simply because of formatting differences.

Other issues required more deliberate analytical decisions. One of the most important tasks was recalculating the readmission flag using the 30-day readmission rule. Instead of relying on the original flag in the dataset, I reconstructed the logic using admission and discharge timestamps to calculate the number of days between encounters for each patient. This ensured that the readmission indicator accurately reflected the standard definition used in healthcare performance analysis.

Another notable issue involved the Pediatrics department, where many records contained patients who were clearly adults based on their age. Rather than reassigning these records to other departments without sufficient evidence, I created a "Misclassified" placeholder. This decision preserves the integrity of the data while clearly signaling that the records may have been routed incorrectly in the source system.

The dataset also contained missing values in the cost of encounter column, which were replaced using the median cost value. The median was chosen because the cost distribution appeared skewed, making it a more robust measure than the mean. In addition, inconsistencies where patients had multiple ages or genders recorded across different encounters were addressed by using the median age per patient and the most frequently occurring gender to create consistent demographic values.

To improve interpretability, coded fields such as ICD-10 diagnosis codes and CPT procedure codes were also mapped to readable descriptions using lookup tables. This makes the dataset easier to interpret for stakeholders who may not be familiar with medical coding systems.

Overall, this cleaning process transformed a dataset that initially contained several inconsistencies into a more structured and analytically reliable resource. The key variables are now standardized, logical conflicts have been addressed, and important derived metrics such as readmission have been recalculated using transparent methods. Importantly, the steps taken were carefully documented to ensure that every change made can be clearly understood and justified.

With these improvements in place, the dataset is now better prepared for meaningful analysis, performance measurement, and dashboard development. Just as importantly, the process reinforced an important lesson: a dataset that looks clean at first glance may still contain hidden issues that only become visible through careful investigation and thoughtful data preparation.